

Details of Field of Research subareas

Numbered codes are the Australian and New Zealand government approved codes. CSE has been added by ICORE to cover this area which lies outside of Information and Computing Sciences in the Australasian system.

4601 Applied computing

- 460101 Applications in arts and humanities
- 460102 Applications in health
- 460103 Applications in life sciences
- 460104 Applications in physical sciences
- 460105 Applications in social sciences and education 460106
- Spatial data and applications

4602 Artificial intelligence

- 460201 Artificial life and complex adaptive systems
- 460202 Autonomous agents and multiagent systems
- 460203 Evolutionary computation
- 460204 Fuzzy computation
- 460205 Intelligent robotics
- 460206 Knowledge representation and reasoning
- 460207 Modelling and simulation
- 460208 Natural language processing
- 460209 Planning and decision making
- 460210 Satisfiability and optimisation
- 460211 Speech production
- 460212 Speech recognition

4603 Computer vision and multimedia computation

- 460301 Active sensing
- 460302 Audio processing
- 460303 Computational imaging
- 460304 Computer vision
- 460305 Image and video coding
- 460306 Image processing
- 460307 Multimodal analysis and synthesis
- 460308 Pattern recognition
- 460309 Video processing

4604 Cybersecurity and privacy

- 460401 Cryptography
- 460402 Data and information privacy
- 460403 Data security and protection
- 460404 Digital forensics
- 460405 Hardware security
- 460406 Software and application security
- 460407 System and network security

4605 Data management and data science

- 460501 Data engineering and data science
- 460502 Data mining and knowledge discovery
- 460503 Data models, storage and indexing
- 460504 Data quality
- 460505 Database systems

- 460506 Graph, social and multimedia data
- 460507 Information extraction and fusion
- 460508 Information retrieval and web search
- 460509 Query processing and optimisation
- 460510 Recommender systems
- 460511 Stream and sensor data
- 4606 Distributed computing and systems software
 - 460601 Cloud computing
 - 460602 Concurrent/parallel systems and technologies
 - 460603 Cyberphysical systems and internet of things
 - 460604 Dependable systems
 - 460605 Distributed systems and algorithms
 - 460606 Energy-efficient computing
 - 460607 High performance computing
 - 460608 Mobile computing
 - 460609 Networking and communications
 - 460610 Operating systems
 - 460611 Performance evaluation
 - 460612 Service oriented computing
- 4607 Graphics, augmented reality and games
 - 460701 Computer aided design
 - 460702 Computer graphics
 - 460703 Entertainment and gaming
 - 460704 Interactive narrative
 - 460705 Procedural content generation
 - 460706 Serious games
 - 460707 Sound and music computing
 - 460708 Virtual and mixed reality
- 4608 Human-centred computing
 - 460801 Accessible computing
 - 460802 Affective computing
 - 460803 Collaborative and social computing
 - 460804 Computing education
 - 460805 Fairness, accountability, transparency, trust and ethics of computer systems
 - 460806 Human-computer interaction
 - 460807 Information visualisation
 - 460808 Mixed initiative and human-in-the-loop
 - 460809 Pervasive computing
 - 460810 Social robotics
- 4611 Machine learning
 - 461101 Adversarial machine learning
 - 461102 Context learning
 - 461103 Deep learning
 - 461104 Neural networks
 - 461105 Reinforcement learning
 - 461106 Semi- and unsupervised learning
- 4612 Software engineering
 - 461201 Automated software engineering
 - 461202 Empirical software engineering
 - 461203 Formal methods for software

461204 Programming languages
461205 Requirements engineering
461206 Software architecture
461207 Software quality, processes and metrics
461208 Software testing, verification and validation

4613 Theory of computation

461301 Coding, information theory and compression
461302 Computational complexity and computability
461303 Computational logic and formal languages
461304 Concurrency theory
461305 Data structures and algorithms
461306 Numerical computation and mathematical software
461307 Quantum computation

CSE Computer Systems Engineering

- Computer Architectures
- VLSI Testing and Dependability
- VLSI Design
- Embedded and Cyber-physical Systems Design
- High Performance Computing
- Architectures for Industrial Informatics
- Robotics Architectures
- Quantum Computing Engineering